

Course Code- 151303**Object Oriented Programming Using Java****3 0 0 3****Unit- 1.0****5 hrs****OOP Concepts and Java Programming:**

Introduction to Java: History of java, Java buzzwords, basics of Java programming, difference between procedural and object-oriented programming paradigm, need for OOPs paradigm, OOPs features, advantages of oops, JDK, JRE and JVM, data types, variables, operators, control structures including selection, looping, java methods, compilation, and execution of simple program.

Unit- 2.0**7 hrs****Objects, Classes, and Constructors in Java:**

Objects and Classes: Basics of objects and classes in java, declaring objects, new keyword, Defining and calling methods in class, Array of Objects, Constructors, Different types of Constructors, Overloading Methods and Constructors, Method Binding, Overriding and Exceptions, Passing Object as parameters, returning object, Static members, Concept of Access Modifiers (Public, Private, Protected, Default), this Keyword, garbage Collection, finalize() method, Nested and Inner Classes, Exploring string class.

Unit- 3.0**9 hrs****Inheritance, Interfaces and Packages:**

Inheritance: Inheritance hierarchies, Benefits of Inheritance, super and subclasses, member access rules, super keyword, preventing inheritance: final classes and methods, the object class and its methods; Polymorphism: Dynamic binding, method overriding, abstract classes and methods; Interface: Interfaces vs Abstract classes, defining an interface, implement interfaces, accessing implementations through interface references, extending interface; Packages: Defining, creating and accessing a package, understanding CLASSPATH, importing packages.

Unit- 4.0**6 hrs****Exception Handling:**

Introduction to error and exception, Error vs Exception, Concepts of Exception Handling, Benefits of exception handling, exception types, exception hierarchy, checked and unchecked exceptions, usage of try, catch, throw, throws and finally, multiple catch clauses, nested try statements, re throwing exceptions, creating own exception sub classes.

Unit- 5.0**7 hrs****Introduction to Multithreading:**

Differences between multiple processes and multiple threads, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, inter thread communication.

Unit- 6.0**8 hrs****Files, The Collections Framework and Connecting To Database:**

Files: Streams, byte streams, character stream, text input/output, binary input/output, random access file operations, file management using file class; The Collections Framework (java.util): Collections overview, Hierarchy of Collection Framework, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque; Connecting to Database: Connecting to a database, querying a database and processing the results, updating data with JDBC.

Text/ Reference:-

1. Schildt, Herbert. "Java: The Complete Reference, Eleventh Edition", New York: McGraw-Hill Education, 2018.
2. Schildt, Herbert. "Java: A Beginner's Guide, Eighth Edition", US: McGraw-Hill Osborne Media, 2018.
3. Balagurusamy, E. "Programming with Java", United States: McGraw-Hill Education, 2019.
4. Herbert Schildt, Dale Skrien. "Java Fundamentals – A Comprehensive Introduction", McGraw-Hill, 1st Edition, 2013.

B. Tech in CSE (AI)**(SESSION- 2024-28 onwards)**

5. P. J. Deitel, H. M. Deitel, “Java: How to Program”, Prentice Hall, 6th Edition, 2005.
6. P. Radha Krishna, “Object Oriented Programming through Java”, Universities Press, CRC Press, 2007.
7. Bruce Eckel, “Thinking in Java”, Prentice Hall, 4th Edition, 2006.
8. Sachin Malhotra, Saurabh Chaudhary, “Programming in Java”, Oxford University Press, 2nd Edition, 2014.

